

PhD Proposal in Biomolecular Modeling

Deep Learning and Molecular Dynamics Simulations Combined to Model Conformational Ensembles of Intrinsically Disordered Proteins and their Complexes

Doctoral School : Innovation Thérapeutique du Fondamental à l'Appliqué (ED IFTA)

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Project description

Nobel Prize-winning AlphaFold deep learning model [1] has proven revolutionary for its success in predictions of protein static structure. However, describing protein dynamics using machine learning methods remains a major challenge, although progress has been made through several approaches. The goal of this project is to combine physics-based molecular dynamics simulations and deep learning models to achieve a fast and reliable description of conformational ensembles for highly dynamic proteins.

It is now well established that intrinsically disordered proteins — flexible proteins that can dynamically adopt different conformations depending on their environment and interactions — are biologically functional. Their broad conformational variability is essential for performing cellular processes [2]. On the other hand, both experimental techniques and molecular dynamics simulations face limitations in characterizing the conformational ensembles of such systems. In particular, molecular dynamics simulations have the advantage to describe protein dynamics but struggle to overcome energy barriers separating distinct conformations. To address these limitations, enhanced sampling methods such as replica exchange [3] or umbrella sampling [4] can be

used, albeit at the expenses of high computational cost. With the advent of machine learning models for single protein structure prediction, new strategies have been proposed to describe conformational ensembles, including modifications to multiple sequence alignments [5, 6]. Mainly inspired by image processing techniques, generative deep learning models such as those based on denoising [7, 8] have also been adapted to generate molecular conformations.

For this project, the candidate will develop novel strategies to sample conformational ensembles using molecular dynamics simulations data to train a deep learning model (FIGURE 1). This model will be designed to generate ensembles starting from protein sequences, but also from structure templates or short molecular dynamics trajectories. This approach will then be extended to model protein complexes and binding interfaces. Comparisons between different models, such as protein language models [9, 10], diffusion-based [11, 12] or flow matching [13, 14], may be carried out to assess convergence and determine the most effective protocol for the objectives of this project.

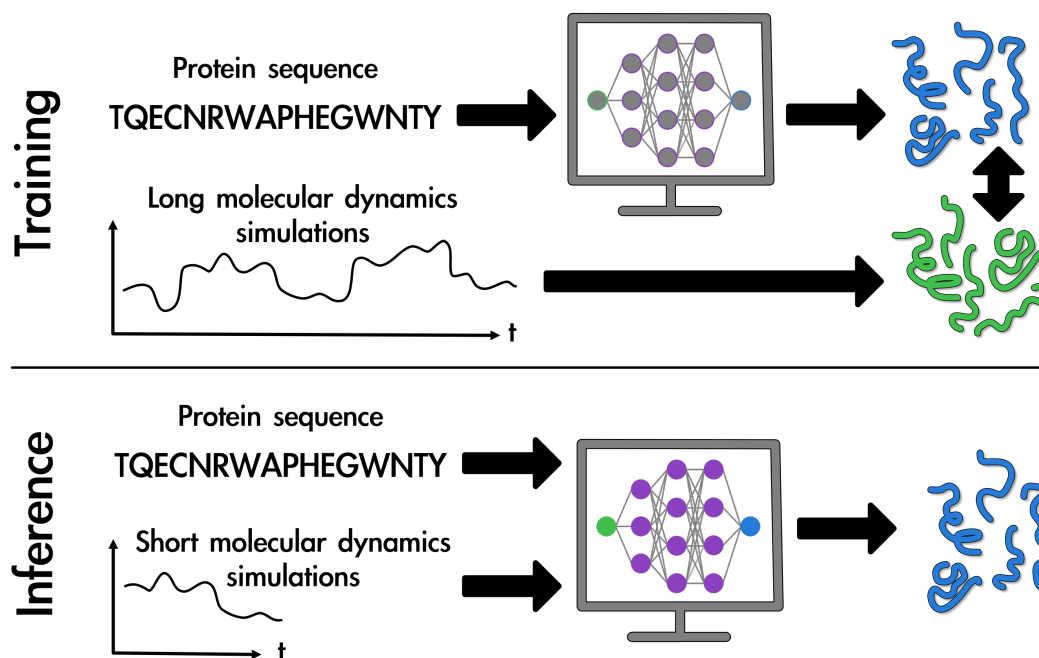


FIGURE 1 – Schematic description of the PhD proposal.

Required skills : Candidates for this thesis project should have a Master 2 (or equivalent) in one or several following scientific fields : data sciences, artificial intelligence, molecular biophysics, structural bioinformatics, medicinal chemistry.

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